

Create

New Graphs

Tree Item
 Choose the parent for this header/graph.
 [Tree]

Tree Item Type
 Choose what type of tree item this is.
 [Host]

Tree Item Value
 [Mihalis' Cisco 877W router (cisco: 1)]

Host
 Choose a host here to add it to the tree.
 [Mihalis' Cisco 877W router (cisco: 1)]

Graph Grouping Style
 Choose how graphs are grouped when drawn for this particular host on the tree.
 [None]

Round Robin Archive
 Choose a round robin archive to control how Graph Thumbnails are displayed when using Tree Export.
 [Hourly (3 Minute Average)]

Cancel **Create**

Figure 4: Creating the graph for the Cisco ADSL router

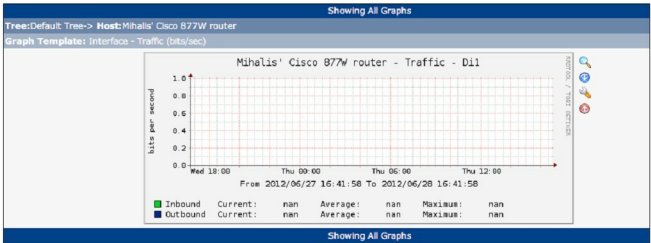


Figure 5: The Cacti graph that was created

- The steps for adding the Cisco router are the following:
- Select 'Devices' on the left.
 - Select Type: 'Cisco Router' and Status 'Enabled' and then click 'Add'.

You will see what's shown in Figure 2, and you will have to fill in the information.

The most important fields are the Hostname ('cisco' in my case), the SNMP Community string ('CactiCom' in my case), and the SNMP Version in which you will have to select 'Version 2'. Then click 'Create'.

On the next screen, you will have to press the 'Create Graphs for this Host' on the upper right side. You will then see Figure 3. This figure lists all the available Cisco interfaces for this particular router. What interests us is Interface No 14 (Dialer1), which is the ADSL Internet connection interface. The desired graph type should be 'In/Out Bytes'.

Select your interface of choice and click the 'Create' button. Another interesting interface is Number 5 (Dot11Radio0), which is Cisco's Wi-Fi interface.

Then, select 'Graph Three' from the left menu and click 'Add'. Select the options that you can see in Figure 4. Then click 'Create'.

Select the 'Graphs' tab and then, from the 'Default Tree', select the desired host. You will have to wait a little, until some data is obtained for the graph to be created. You will see

something similar to Figure 5, as the graph will take a little while to get populated. If you click on the graph, you will get daily, weekly, monthly and yearly graphs.

The 'Graph' tab is the key screen for end users to view the graphs for their devices and to change settings.

Cacti is a very capable tool but in order to harness its power you need to experiment with it. An easier tool to configure is MRTG, but it has less capabilities.

Cacti also allows you to use existing plug-ins or create your own. Plug-ins allow developers to generate additional Cacti features without changing Cacti's source code. I strongly recommend learning both Cacti and MRTG in order to use the right tool for the right job. 

Web links and bibliography

- [1] Cacti: <http://www.cacti.net/>
- [2] Cisco MIBs: <http://www.cisco.com/public/sw-center/netmgmt/cml/mibs.shtml>
- [3] SNMP RFCs: http://www.snmp.org/protocol/snmp_rfc.shtml
- [4] RRDtool: <http://oss.oetiker.ch/rrdtool/>
- [5] MRTG: <http://oss.oetiker.ch/mrtg/>
- [6] Cacti plug-ins: <http://docs.cacti.net/plugins>

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